Homework 2 Seguridad y Criptografía

Course code:	MC1003
Due:	October 28, 2026 at 23:59 (Central Standard Time)
Total points:	10 points

All answers need to be motivated (unless stated otherwise), i.e.*, just stating the answer is not enough.*

1. **One-time Pad** *(2 points)*

   Mallory intercepts an encrypted (hexadecimal encoded) message,

   $$\mathsf{ct} = \text{3b30cf656ed9fe4df54e08bc57e8f247c654977b7decfa9f2c171b},$$

   from Alice to Bob. Mallory knows that Alice and Bob communicate using the one-time pad and she is sure the underlying plaintext starts with "Trust Charlie".

   How can Mallory change the message ct to ct' in such a way that the decryption of ct' starts with "Trust Mallory"?

2. **Linear Feedback Shift Register (LFSR)** *(4 points)*

   For a 3-bit LFSR, write the register as $s_n s_{n+1} s_{n+2}$, and number the three positions $0, 1, 2$ from left to right. This means that the new bit is the bit three positions back. A *tap set* $T \subseteq \{0, 1, 2\}$ means that the next bit is computed as $s_{n+3} = \bigoplus_{i \in T} s_{n+i}$, where $\oplus$ is XOR (addition modulo 2). The output sequence is $s_0, s_1, s_2, \ldots$.

   (a) (2 points) Given the tap set $T = \{0\}$ for a three-bit LFSR, gGive (at least 8 bits of) the output sequence of the LFSR for the initial state $s = s_0 s_1 s_2 = 0\,1\,0$.

   (b) (2 points) Given the output

   $$s = 1, 1, 1, 0, 0, 1, 0, 1, \ldots$$

   of a three-bit LFSR. Determine the tap set T.

3. **Linear Complexity of an LFSR** *(4 points)*

   Let $s = s_0, s_1, \ldots$ and $t = t_0, t_1, \ldots$ be two sequences. The *linear complexity* $L(s)$ is the length of the shortest LFSR that can produce the sequence s. According to the textbook (*Cryptography Made Simple*), it satisfies

   $$L(s \oplus t) \leq L(s) + L(t),$$

   where $s \oplus t$ means bitwise XOR of the two sequences.

   (a) (1 point) Give an example of two sequences s and t, such that

   $$L(s \oplus t) < \min(L(s), L(t)).$$

   (b) (3 points) Give an example of two sequences s and t, such that

   $$L(s \oplus t) = L(s) + L(t).$$